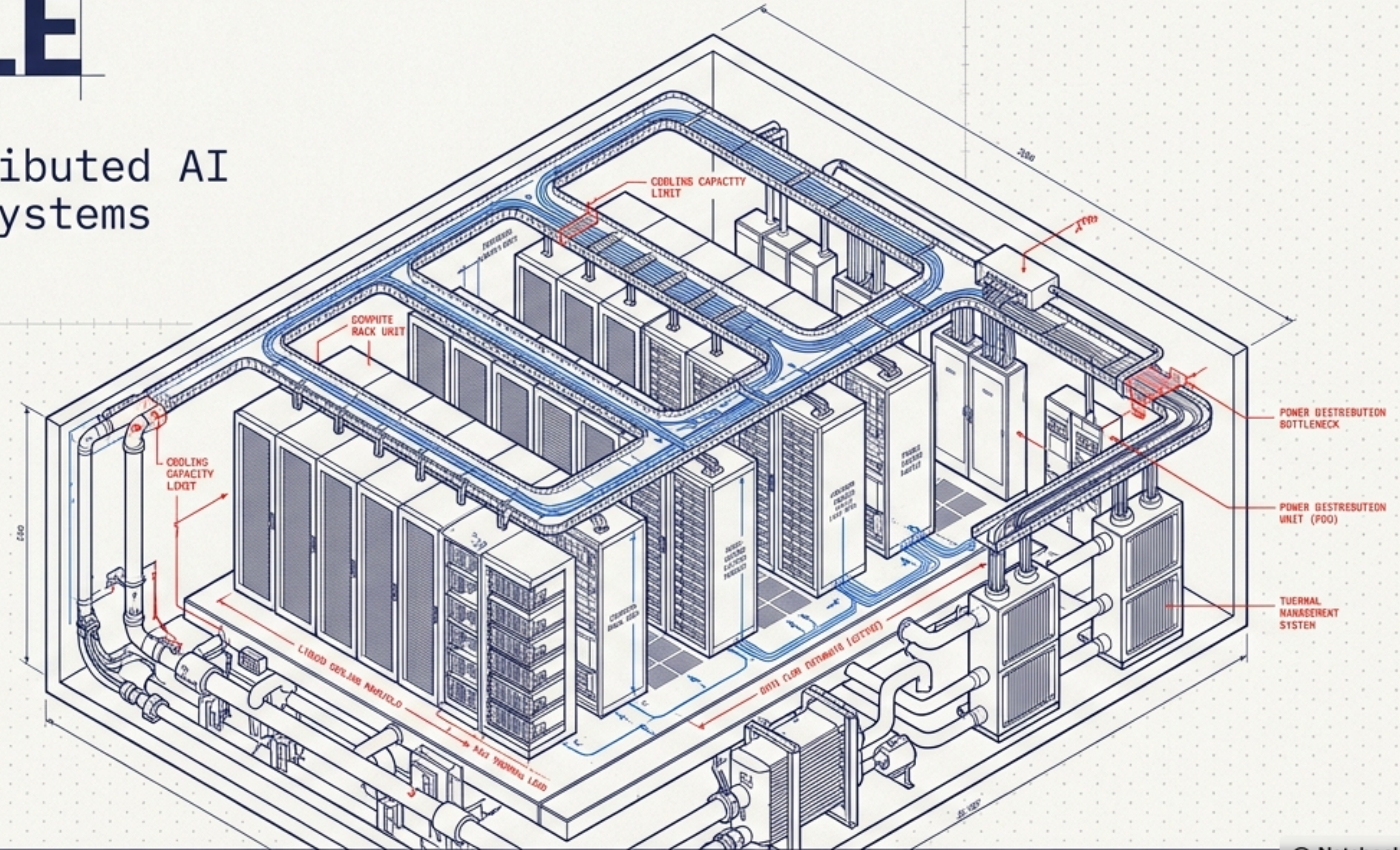


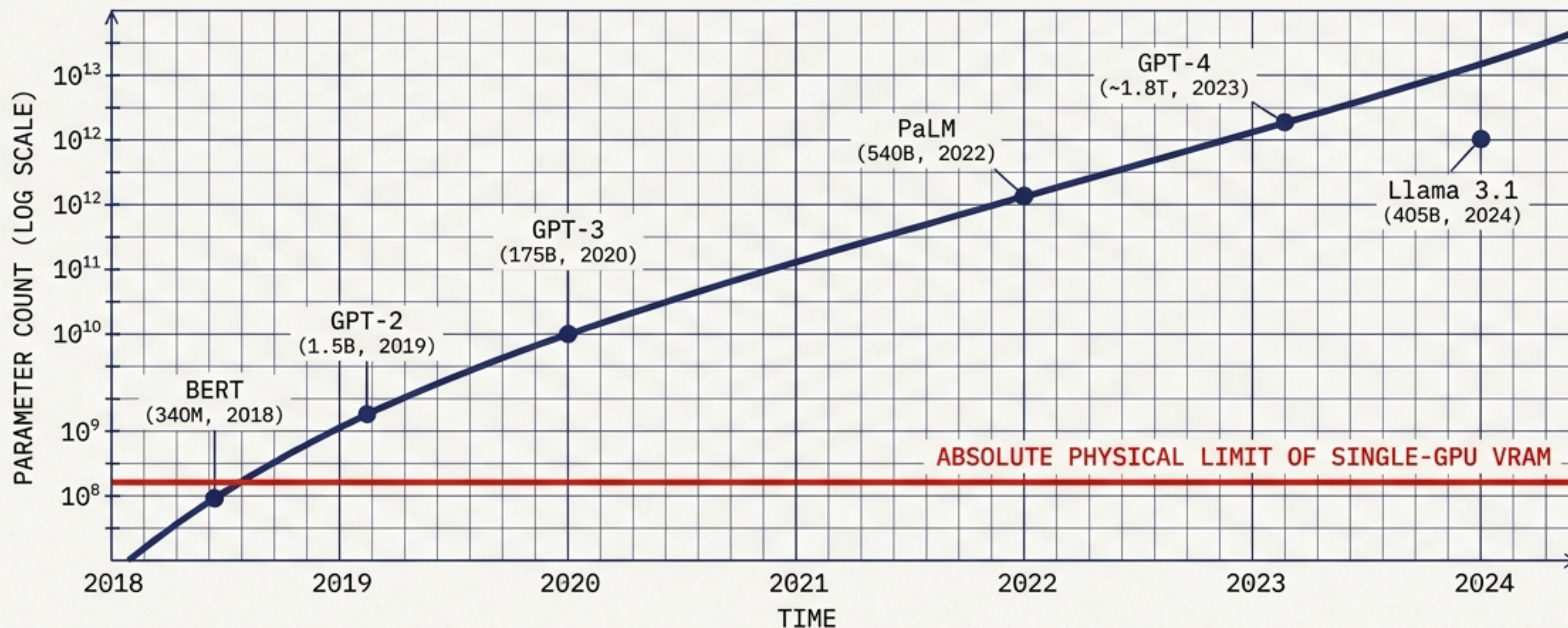
# THE ARCHITECTURE OF SCALE

Engineering Distributed AI  
from Physics to Systems





# THE PHYSICS OF SCALE DICTATE STRUCTURAL CHANGE



## ENGINEER'S MAXIM

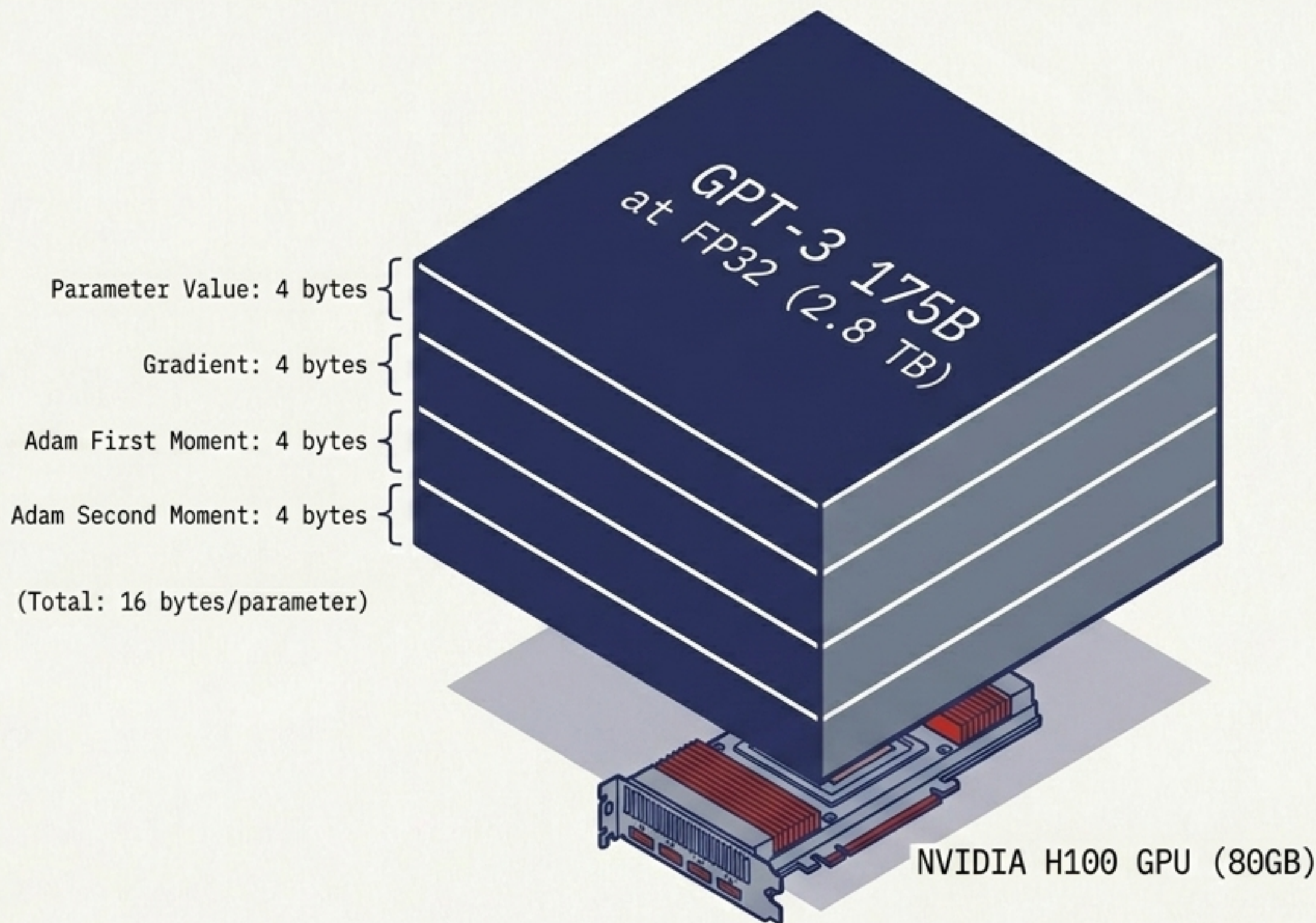
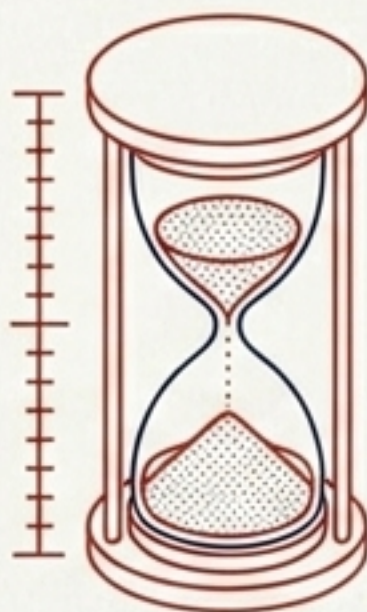
Any engineer expecting a bigger single machine to eventually solve large-scale AI training is operating under a category error. Distributed training is the permanent structural answer.



# THE SINGLE-MACHINE COLLAPSE: A PROBLEM OF MATHEMATICAL VOLUME

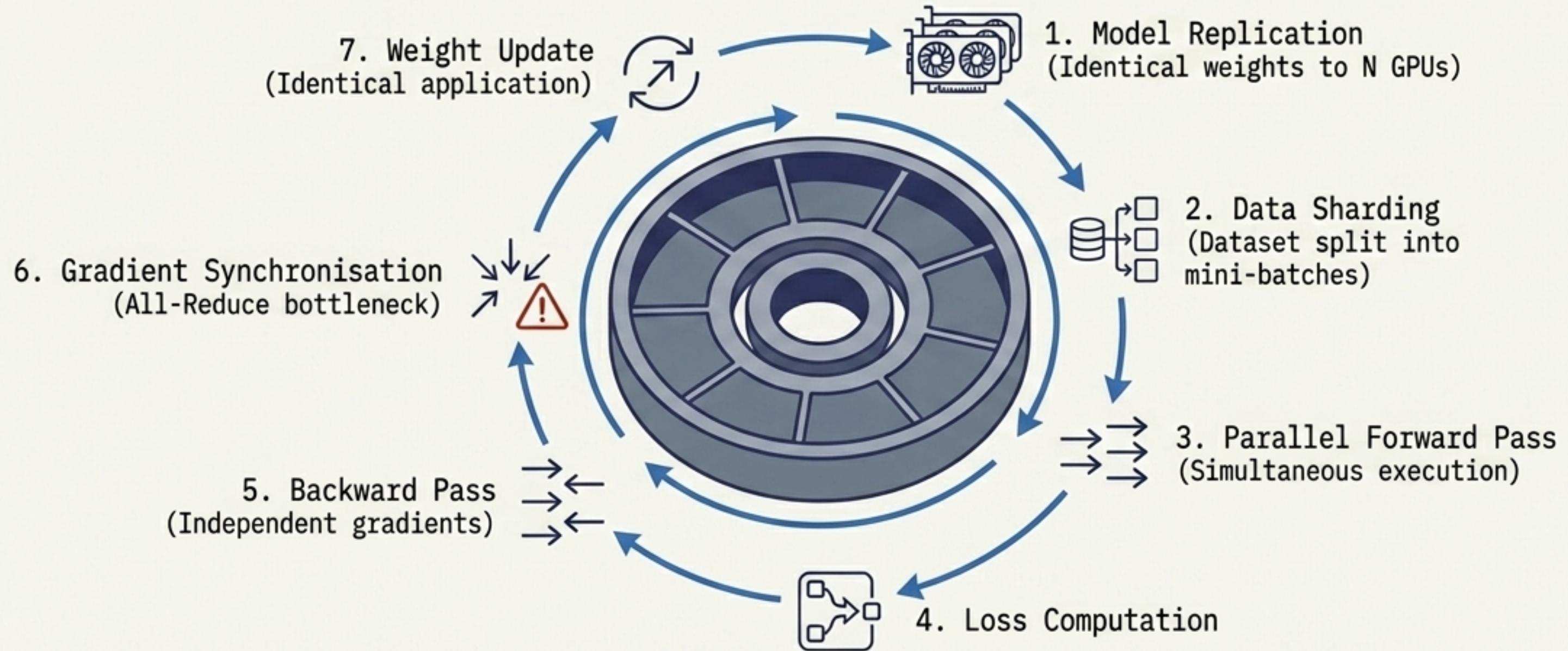
## COMPUTE THROUGHPUT REALITY

Even if a 2.8TB model could fit into a single GPU's memory, processing 1 trillion tokens at 2,000 TFLOPs (theoretical H100 peak) would take one machine 1,585 years.





# 1D: THE DATA PARALLELISM ENGINE

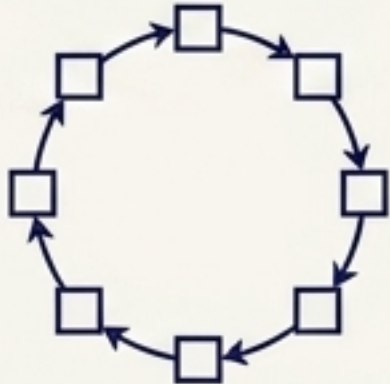
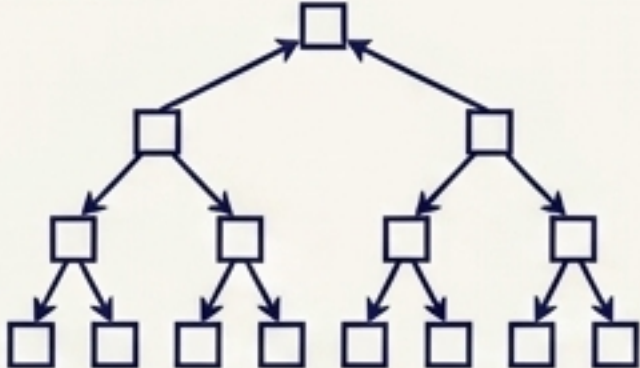
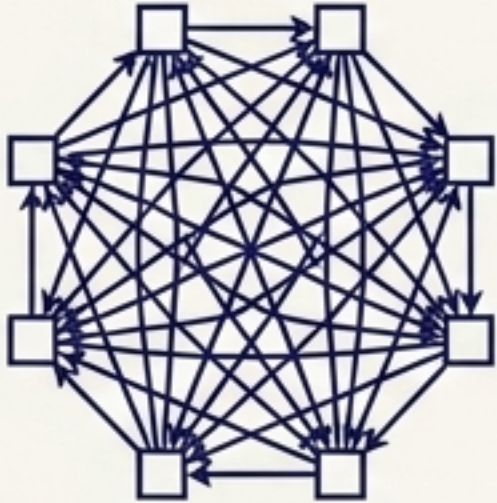


## ENGINEER'S MAXIM: THE LINEAR SCALING RULE

When expanding your cluster to N GPUs, your effective batch size becomes  $N \times B$ . You must increase your learning rate by the exact same factor to maintain convergence dynamics.



# SYNCHRONISATION TOPOLOGIES: THE ALL-REDUCE MATRIX

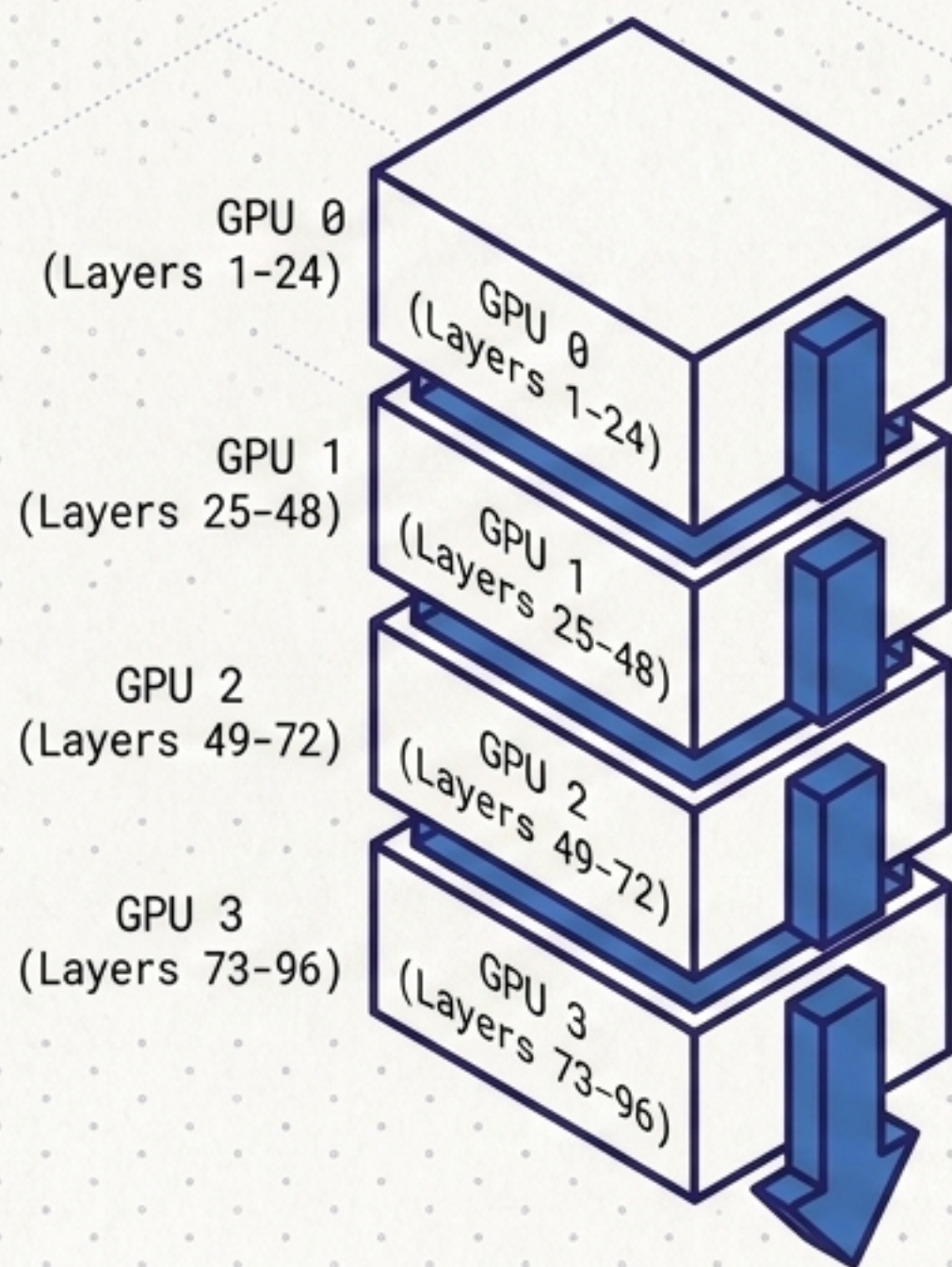
| ALGORITHM       | TOPOLOGY SCHEMATIC                                                                    | LATENCY PROFILE                               | IDEAL APPLICATION                       |
|-----------------|---------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------|
| Ring All-Reduce |    | $O(N)$ latency,<br>optimal bandwidth          | Large clusters,<br>bandwidth efficiency |
| Tree All-Reduce |    | $O(\log N)$ latency,<br>high bandwidth use    | Small clusters<br>requiring low latency |
| NCCL All-Reduce |  | Hardware-optimised,<br>near-theoretical speed | NVLink-connected<br>GPU pods            |

DATA BOTTLENECK CONSTRAINT: Synchronising a 175B parameter FP32 model requires transmitting ~700GB of data per step over the interconnect fabric.

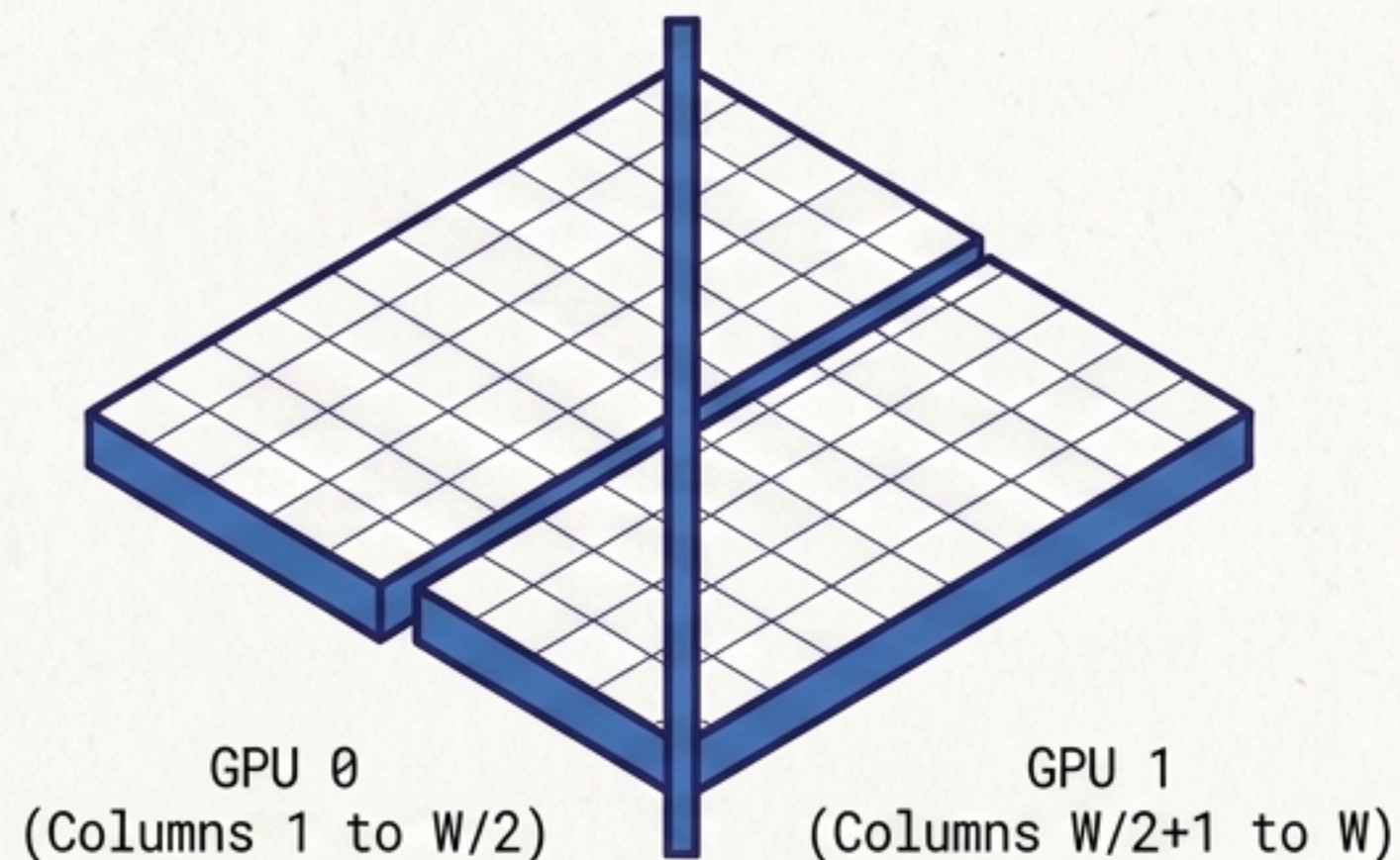


# 2D: THE ANATOMY OF MODEL PARALLELISM

Pipeline Parallelism  
(Horizontal Slicing)



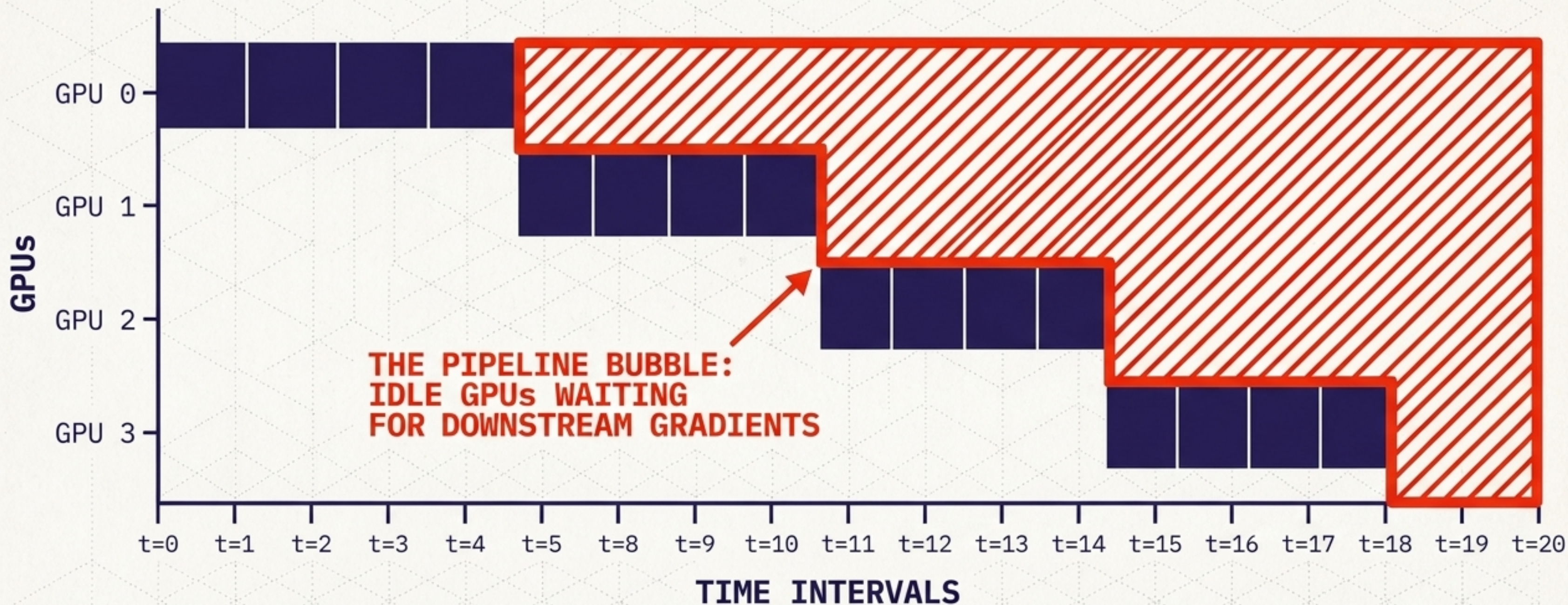
Tensor Parallelism  
(Vertical Slicing)



Intra-layer partitioning. Math operations execute simultaneously, demanding massive NVLink bandwidth to instantly merge results.



# THE PIPELINE BUBBLE: DIAGNOSING COMPUTE WASTE



**DIAGNOSTIC DATA POINT:** In a naive 4-stage pipeline without micro-batch interleaving, up to 75% of theoretical compute time is lost to this bubble overhead.



# THE PARALLELISM DIAGNOSTIC ENGINE

## 1D: Data Parallelism

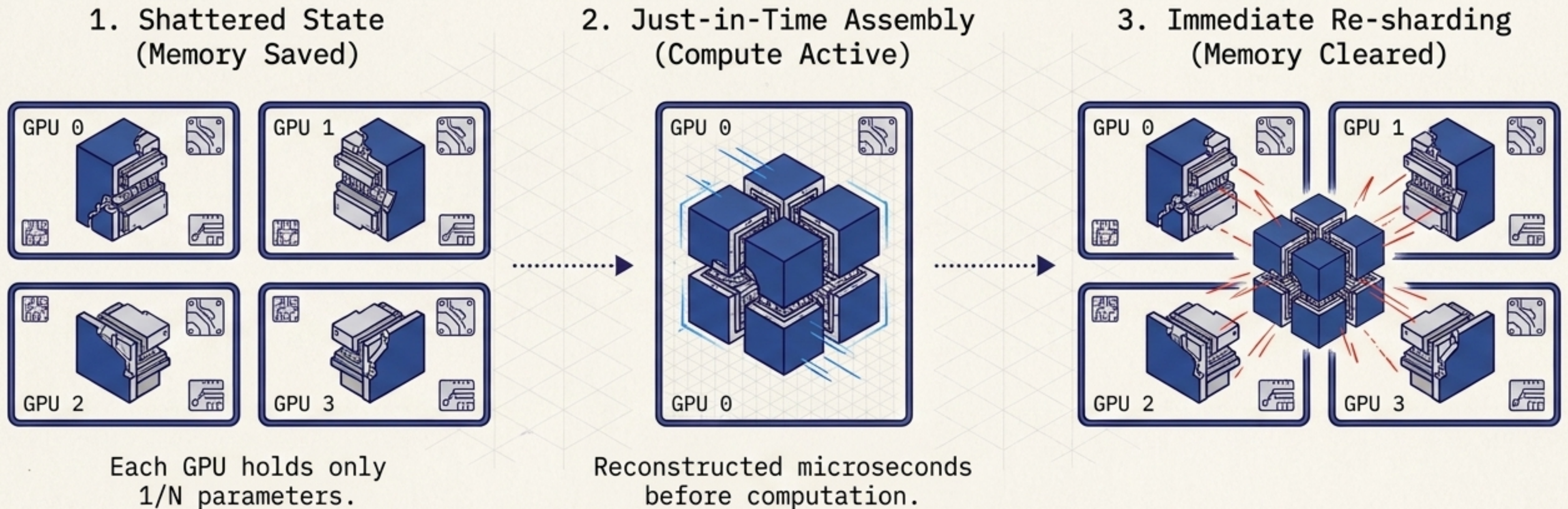
-  **Model Footprint:** Full model replicated on every GPU.
-  **Data Flow:** Training dataset is split across the GPUs.
-  **Comms Frequency:** Lower (synchronises once per step).
-  **Throughput:** Near-linear scaling.
-  **Trigger Condition:** Use when model fits in a single GPU and maximum throughput is required.

## 2D: Model Parallelism

-  **Model Footprint:** Physical model architecture is split across GPUs.
-  **Data Flow:** The exact same data flows through all devices sequentially.
-  **Comms Frequency:** High (continuous passing of activations and gradients).
-  **Throughput:** Sub-linear scaling (restricted by pipeline/tensor overhead).
-  **Trigger Condition:** Mandatory when the model exceeds the 80GB VRAM barrier.



# FSDP: The Hybrid Lazy Assembly Sequence

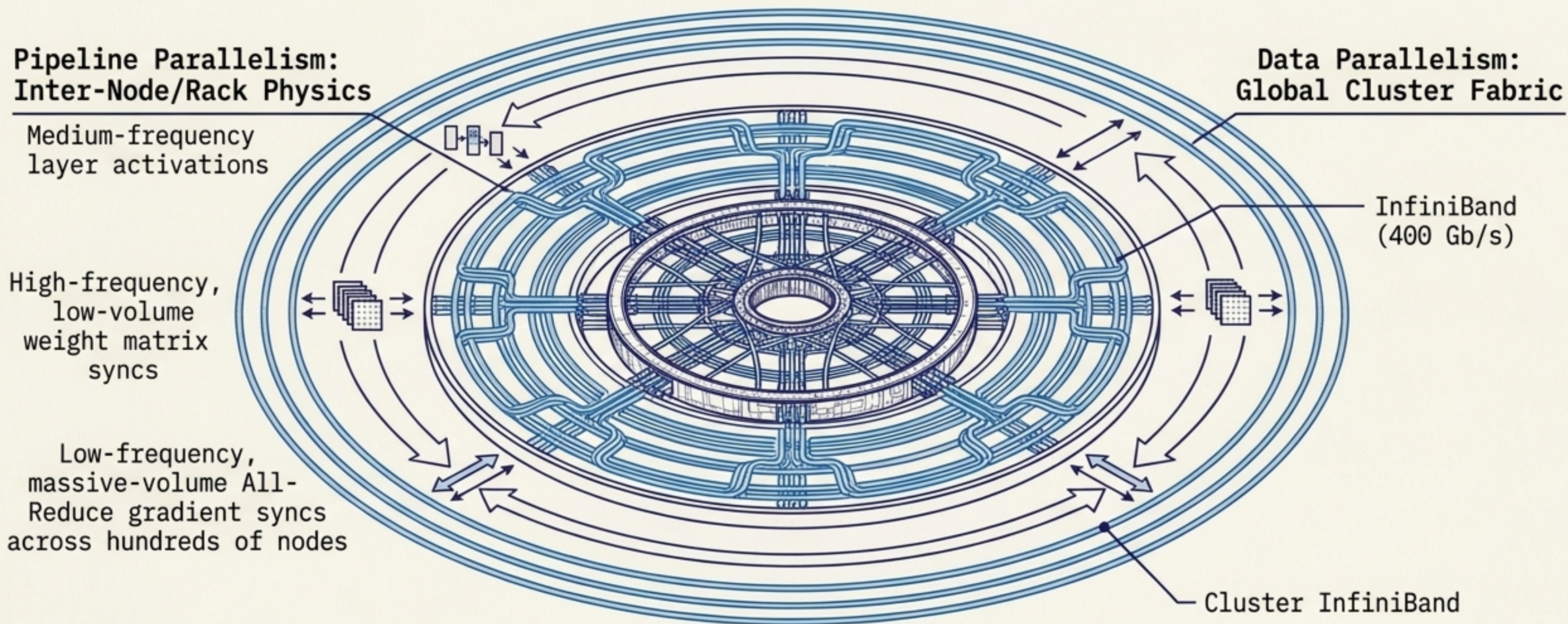


## The Parameter Sharding Spectrum

|  |                                                                                           |
|--|-------------------------------------------------------------------------------------------|
|  | <b>DDP:</b> Full model per GPU. Full gradient. Full optimizer.                            |
|  | <b>FSDP:</b> 1/N of model. 1/N of gradients. 1/N of optimizer. (Ideal for 7B-70B models). |
|  | <b>DeepSpeed ZeRO:</b> 1/N of everything, fully partitioned. (Ideal for 100B+ models).    |



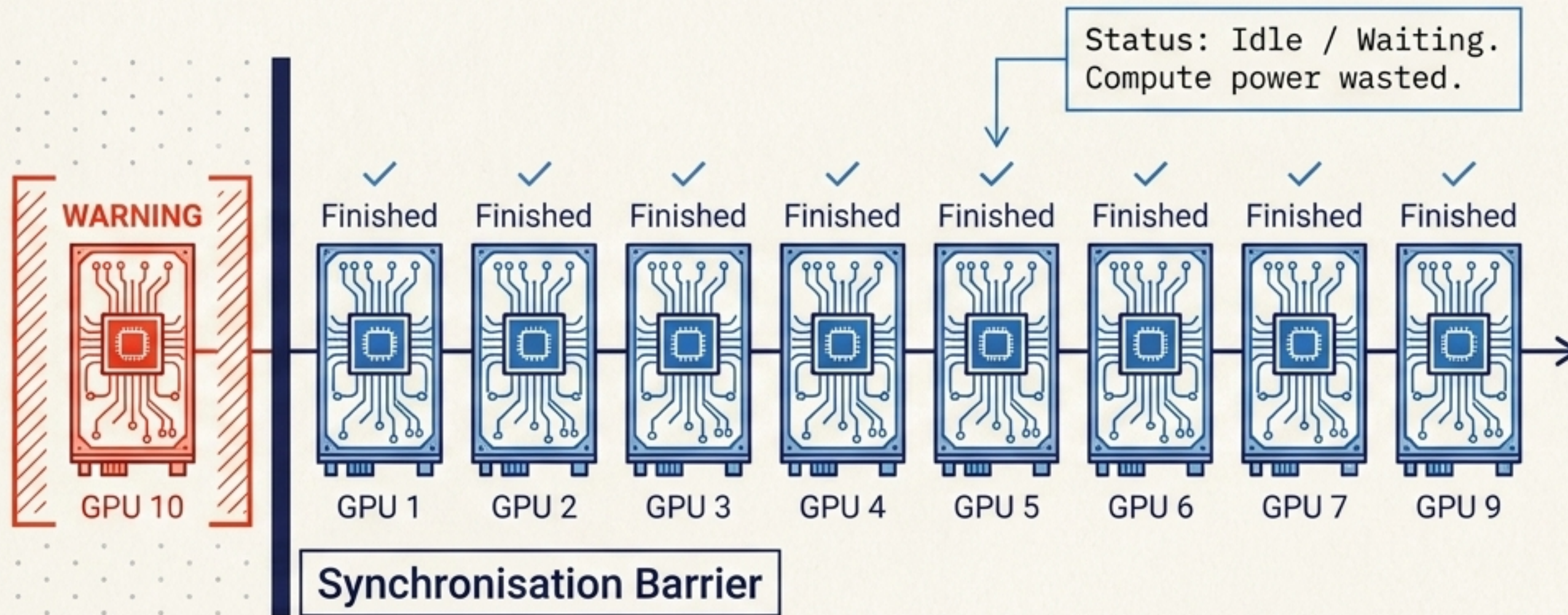
# The 3D Parallelism Topology Map



**Core Insight:** Trillion-parameter training (1024+ GPUs) requires all three layers operating simultaneously. Software orchestration perfectly mirrors hardware networking constraints.



# The Friction of Scale: The Straggler Ripple Effect



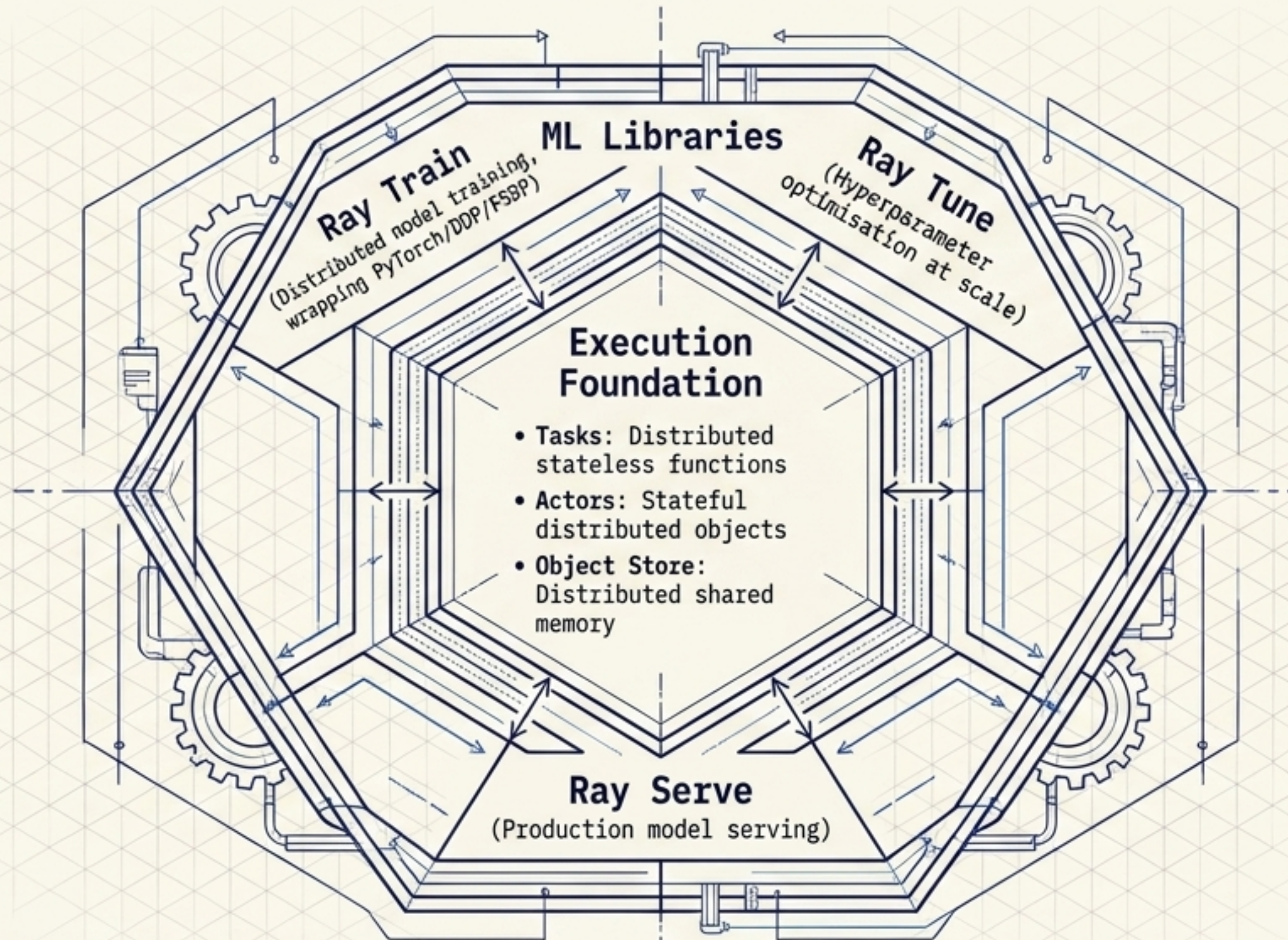
**Engineer's Maxim: One 10% slowdown  
degrades 100% of the cluster.**

## Operational Reality

- At 1,000 GPUs with a 1-year MTBF, expect ~2.7 GPU hardware failures per day.
- Rule of Thumb: For 500+ GPU runs, checkpoint to redundant storage every 10 minutes.
- Mixed-precision checkpoints (FP32 master, FP16 training) are mandatory.



# The Orchestration Engine: The Ray Ecosystem



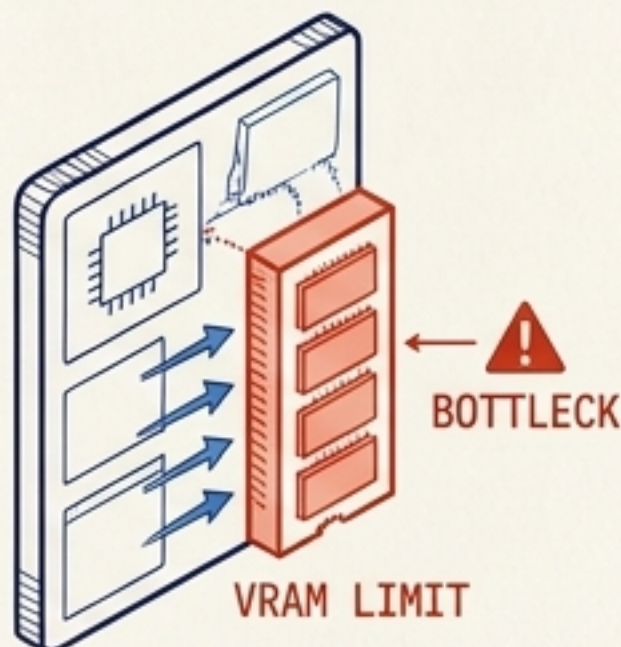
## Core Philosophy: Write Once, Run Anywhere.

Ray transforms standard Python functions into massively scalable distributed objects, abstracting away cluster management, failure recovery, and network scheduling.



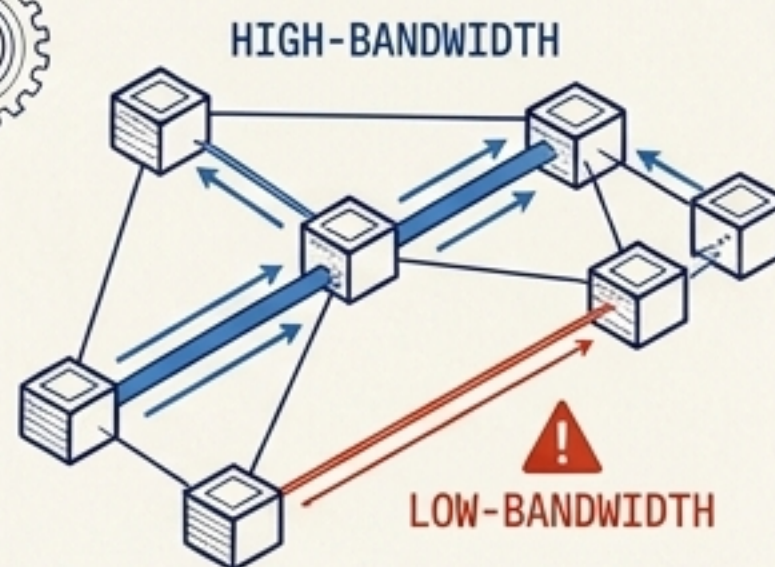
# The Distributed Engineer's Manifesto

## I. Compute is a Physics Problem



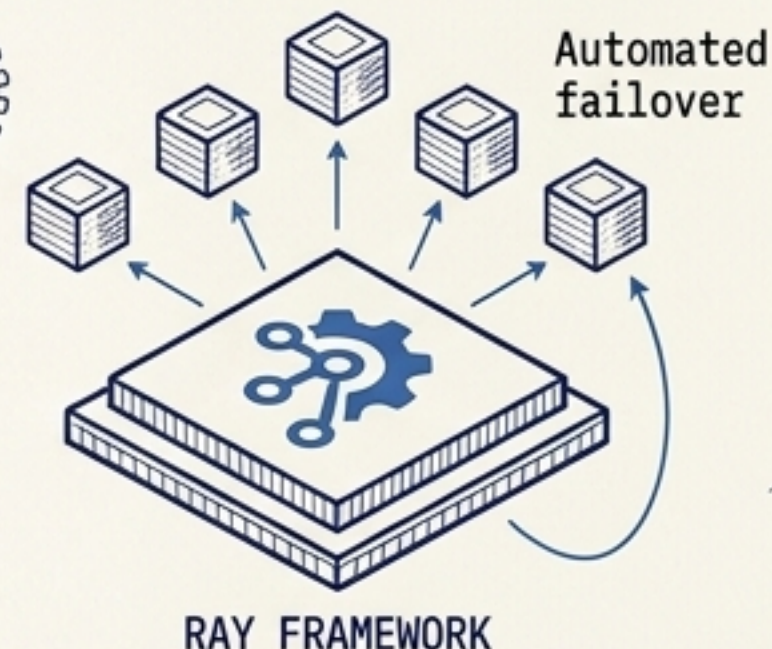
Scale is strictly limited by VRAM. Address memory constraints first (using FSDP or Tensor parallelism) before optimising for raw throughput.

## II. Communication is the Bottleneck



The architecture of your software must map perfectly to the physical topology of your network. Do not run high-volume syncs across low-bandwidth links.

## III. Infrastructure Determines Velocity



Manual cluster management at scale is a dead end. Foundational frameworks like Ray are absolutely required to abstract node failure and cluster elasticity.

**The Single-Machine Era is permanently over. Scale is the new default.**